European Commission

EU Missions assessment report

Mission Restore our Ocean and Waters

Independent Expert Report

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EU Missions assessment report: Restore our Ocean and Waters

EXECUTIVE SUMMARY

THE MISSION RESTORE OUR OCEAN AND WATERS: A FIRST ASSESSMENT OF PROGRESS

The assessment of the EU Mission Restore our Ocean and Waters by 2030 was carried out between November 2022 and May 2023. The assessment examined four dimensions: the selection process; governance structures and management arrangements; budget and funding for implementation and progress towards meeting the mission's goals. It provides options for the future development of the mission. The assessment is based on quantitative and qualitative evidence collected through primary (interviews, a survey and an online policy workshop) and secondary (mission-relevant studies, policy reports and grey literature; a literature review of academic articles, data on Horizon Europe mission calls and projects, etc.) research.

1. The Mission Restore our Ocean and Waters: from launch to first deployment stage

A mission board of 15 high-level experts started working in autumn 2019 and used background studies and broad consultations with stakeholders and citizens to propose a five-legged mission inspired by the 'starfish' metaphor. The board outlined the fundamental importance of the water system and the need for a systemic and comprehensive approach for its restoration. The mission board report was used as a basis by the Commission services to develop the Mission Restore our Ocean and Waters. The mission has three specific objectives: 1) protect and restore marine and freshwater ecosystems and biodiversity; 2) prevent and eliminate pollution of our ocean, seas and waters; and 3) make the sustainable blue economy carbon-neutral and circular, supported by two enablers: 1) a digital ocean and waters knowledge system; and 2) broad public mobilisation. Nine quantitative targets were set across the three objectives.

Until 2030, the mission is to be implemented in two phases: 1) development and piloting phase (2022-2025) and 2) deployment and upscaling phase (2026-2030). The first phase serves to provide the basis for the implementation of the mission objectives and enabling actions. Namely, new transformative solutions for ecosystem restoration, reducing pollution and developing circular and carbon neutral blue economy will be tried and tested.

Four mission 'lighthouses' have been launched as sites to pilot, demonstrate and deploy the mission solutions across the major EU sea and river basins: Atlantic-Arctic, Mediterranean Sea, Baltic-North Sea, and Danube River–Black Sea basin. The lighthouses will catalyse the implementation of mission objectives by strengthening basin-scale cooperation and governance, including stakeholder ownership and citizen engagement.

2. Assessment of the mission

2.1. ASSESSMENT OF SELECTION PROCESS AND SCOPE

Addressing the hydrosphere as a connected system of ocean, seas, coastal and inland waters is ambitious but necessary in the context of the global challenge. As a system-wide framework based on existing legislative and regulatory measures, bringing together different initiatives connected to the entire water lifecycle, the mission goes far beyond the scope of existing initiatives or instruments at EU level. This naturally introduces a high level of complexity into the mission's setup and operations.

The mission's selection process has been transparent, with many opportunities for co-creation and stakeholder engagement. This included a number of citizen initiatives injecting bottom-up support and societal buy-in into the mission processes. Overall, there is confidence that mission scope and objectives are strongly endorsed by the wider public.

2.2. ASSESSMENT OF GOVERNANCE STRUCTURES AND MANAGEMENT ARRANGEMENTS

The mission is closely linked to a range of European and international policy and regulatory initiatives, guided at the EU level by the Green Deal. This poses a significant coordination challenge. The established mission governance model has been an important step forward in carrying out the cross-cutting mission steering work. In this respect, the progress with the joint management of the mission by the Directorates General MARE and RTD regarded as a promising start. While some horizontal coordination efforts work well, other aspects are hindered due to diverging policy approaches or conflicting priorities among the 14 involved DGs. Given the number of DGs involved it is expected that horizontal coordination will remain very challenging. Higher level political steering, e.g. from the European Council, would help to underline the political priorities of the mission and support better alignment of action plans at cross-departmental level.

The four basin-level lighthouses represent the main mechanism for the vertical governance coordination nationally, regionally and locally. At basin level, there are overlapping mandates and activities at different organisational levels. The lighthouse implementation projects seek to address this by providing a 'structuring effect' to better align existing basin-level ecosystem actors. Continued work is needed to ensure that Member States are onboard, especially in those where little interest in or commitment to the mission has been shown to date.

2.3. ASSESSMENT OF THE INTERVENTION LOGIC

The intervention logic foresees multiple change mechanisms that showcase the mission's added value in terms of individual support initiatives. These include mobilising myriad support instruments, strengthening existing and forging new cooperation networks among diverse actors, and implementing lighthouses as place-based multi-level coordination and implementation hubs. This is regarded as a logical and convincing approach which draws on pre-existing stakeholder networks and communities to address all topics from the outset. Nevertheless, some misconceptions still prevail about the lighthouses. The information on how the trans-basin approach is implemented in practice has not yet reached all actors responsible for implementing the mission. Progressing from the development and piloting phase to the deployment and upscaling phase is a challenge ahead.

Overall, there is a partial roadmap on how the resources and activities are expected to accomplish the mission. Explicit connections between resources, activities, outputs, outcomes and impacts are yet to be described. This is perhaps understandable due to the bottom-up, solution-focused framing of the mission's actions rather than top-down edicts charting its directions. Once the strategic commitments are reached at lighthouse level, more scrutiny will be necessary to account for and link the order and scale of the mission's expected outcomes, to ensure a clear and understandable logical pathway describing how mission activities are contributing to the targets. A clearer intervention logic would give added legitimacy and ensure the mission is perceived as a goal-oriented coordination device, not separate clusters of initiatives.

2.4. ASSESSMENT OF THE PORTFOLIO OF INSTRUMENTS AND FUNDING

The mission mobilises diverse policy instruments and actions towards balancing the resources supporting research, innovation and the deployment of solutions, physical and digital infrastructure investments, education, training and skills development, regional investments and local capacity-building, and financial and advisory services. There are clear examples of an ongoing brokering of 'synergy arrangements' at the heart of this mission-oriented approach. In terms of research, innovation and implementation actions, the portfolio of mission instruments and actions is assessed as 'fully fledged' with a long list of ongoing activities and those that are planned or promised. With respect to financial mechanisms, the mission portfolio already includes a substantial EU-level instrument and further support mechanisms are being elaborated.

Opportunities with the financial institutions at national and macro-regional level and philanthropic funds are gradually being identified. Complementary actions include a portfolio of individual and collective actions as well as governance arrangements already initiated or implemented. Overall, the scale of funding available at the EU level appears to be appropriate for undertaking the broad range of the envisioned actions. The partial mapping of current appropriations across many instruments indicates that EU contributions to the mission (direct and indirect) may eventually reach tens of billions of euros per year. However, it will be necessary to mobilise significantly greater funding from national, regional and private sources for the mission's implementation.

2.5. ASSESSMENT OF PROGRESS TOWARDS MEETING THE MISSION'S GOAL

While still in the early stages of implementation, efforts to develop connections between initiatives is regarded as a key achievement by the mission to date. This includes a new paradigm for implementing R&I, development, and piloting projects through Horizon Europe mission calls. The portfolio of other EU supported actions relevant to the mission is being identified and links to mission objectives established. Pledges to the mission charter continue to increase (480 endorsements at the time of writing), translating into new initiatives valued at around €3.72 billion. A mission implementation platform was launched in 2022 as a hub for mission partners, and to host monitoring and progress tracking system.

The mission's goals are measurable and time bound, yet the criteria for achieving them by 2030 are less well defined. The targets are judged very ambitious (even extremely so), but they are aligned with the European Green Deal and reflect the urgency and scale of the challenge. Despite the awareness that not all goals may be reached by 2030, there are important achievements in terms of the joined-up policies, instruments and solutions that put Europe on an accelerated track to achieving most of the outlined goals.

3. Options for the mission's future development

3.1. CONTINUE RAISING THE VISIBILITY OF THE MISSION

There is an overarching message that stakeholders at EU, national and regional levels do not have sufficient understanding of all the different activities linked to mission (going beyond the Horizon Europe mission calls). This holds valid not only for organisations at the regional and national level, but also key stakeholders at the European level. While important communication efforts have already been undertaken in the first years of the mission implementation, most notably by the European Commission (EC) and the mission board, the core messages need significant amplification across a very wide range of communities and communication channels.

3.2. IMPROVING TRACTION AT THE NATIONAL/REGIONAL/LOCAL LEVEL

The key challenge for the mission lies in coordinating and aligning resources across national, regional, and local governance levels. This is the level where all critical activities need to take place, like creating awareness, forging joined-up thinking about solutions, implementing legislation and rolling out support instruments and providing feedback on what is working or not in practice. The awareness about the mission initiative is growing at national and regional levels, yet more efforts are needed to improve the traction at all these levels. Thus far, there has been ambiguity surrounding the political ownership of the mission at the EU level. Firstly, a strong EU commitment to restoring the health of oceans and waters must be signalled, demonstrating unified support from top European decision-making levels. This will underscore the mission's importance as a European priority for the upcoming decade and bridging political cycles.

Secondly, promoting national prioritisation of mission actions is crucial. Member States vary in their response to the mission, requiring effective connections between EU-level steering efforts and the political levers at the national level. There are Member States that have actively embraced the mission initiative, while others are still striving to comprehend the interconnections between this EC initiative and existing ongoing instruments. Thus, offering technical support as a tool for institutional and administrative reforms will facilitate the governance transition. Lastly, highlighting practical synergies at the regional level is essential. By showcasing real-life examples of how champion regions have used existing funding opportunities synergistically, a tangible inspiration for mission-oriented thinking can be fostered. This complements existing conceptual guidance materials and encourages wider efforts in regional contexts.

3.3. BRING INDUSTRY ON BOARD

In the first years of the mission's implementation there has not been a strong focus on engaging industry. Companies find it challenging to understand the EU mission and their potential roles within it. However, businesses are a crucial target group, particularly during the deployment and upscaling phase where new business models are needed to address water system challenges.

To address this issue, it is important to tap into industry networks and generate interest in the opportunities offered by the mission initiative. One suggested approach from the study results is to establish close links between the mission process and industry-oriented partnerships. Involving industry partners in the governance structures of lighthouse projects can ensure relevant messages are effectively shared within business communities. Additionally, engaging clusters and innovation ecosystems in lighthouse activities can help establish the necessary connections.

Another strategy is to leverage successes from collaborations with the European Investment Bank (EIB) and BlueInvest. These partnerships provide risk capital financing to support the mission and target a dynamic ecosystem of companies expected to deliver innovative technologies, solutions, and business models. Showcasing proofs-of-concept and successful pilot projects supported by these collaborations would demonstrate industry's active participation in realising the mission's ambitions.

3.4. CONSOLIDATE A CLEARER ROADMAP TOWARDS REACHING MISSION'S OBJECTIVES

The essence of the mission is to establish new ways of working that mobilises actions in a bottom-up manner while ensuring alignment and direction through top-down guidance. This entails the need for a higher-level strategic roadmap on how resources and activities are expected to lead to the desired outcomes while leaving the detailed pathways towards solutions open-ended. Such a roadmap would give a greater legitimacy to the mission as a concerted endeavour and would also help to communicate it to stakeholders in the future.

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